

MGT 816 • The Private Firm CFO — Session 3

S03 Legal & People Infrastructure — Exercise Solutions

Posted after class

Mon Sep 14, 2026 • Prof. John Barrios • Yale SOM

Exercise: after-tax cash, LLC vs. C-corp

One business, \$100 of pre-tax profit, fully distributed this year. Two legal forms. Where does the owner end up?

Task: Compute owner's after-tax cash under (a) LLC pass-through and (b) C-corp — pay corporate tax, then dividend-tax the remainder.

Time: 3 min solo · 2 min pair · calculator / phone out.

Poll: Which form wins, and by how much per \$100?

A) LLC by <\$5 B) LLC by ≥\$5 C) C-corp wins D) tie

Tax rates for this exercise

Ordinary $t_{\text{ord}} = 37\%$

Corporate $t_{\text{c}} = 21\%$

Shareholder $t_{\text{s}} = 23.8\%$

$t_{\text{s}} = 20\% \text{ capital-gain} + 3.8\% \text{ NIIT.}$

Answer: after-tax cash + the general rule ★

Worked solution

LLC — one layer (pass-through)

\$100 – \$37 (t_{ord} 37%)

= \$63.00

C-corp — two layers

\$100 – \$21 = \$79.00 (t_c 21%)

dividend 23.8% × \$79 = \$18.80

= \$60.20

LLC wins by \$2.80 / \$100 (poll A) — combined C-corp rate 39.8% vs. 37%

General rule: choose pass-through iff $(1 - t_{ord}) > (1 - t_c)(1 - t_s)$.

Indifference: $t_s^* = 1 - (1 - t_{ord}) / (1 - t_c) = 20.3\%$.

The shareholder layer is deferred until distribution/sale — retention shrinks the C-corp penalty.

The misclassification bill ★

10 × \$80,000 × 2 yrs — reduced \$3509 rates (1099s filed)

Your startup paid contractors 10 × \$80,000 × 2 yrs — wages \$1,600,000, 20 worker-years. An IRS/state audit reclassifies them all as W-2. Tally the federal bill.

Line item	Amount
Employer FICA (7.65% × wages)	\$122,400
Employee-FICA share (1.53%)	\$24,480
Income-tax withholding (1.5%)	\$24,000
FUTA (\$420 × 20 wkr-yrs)	\$8,400

Federal bill
\$179,280
≈ 11.2% of wages

≈ 11% of every contractor dollar, before interest, state UI, benefits — rates DOUBLE if no 1099s were filed; ~100% exposure if willful (+ officer liability).

Separate FLSA exposure — misclassified non-exempt workers add 2–3 yrs of back overtime × 2 liquidated damages.